

Praveen Sharma

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. Data Science / B.S. Applied Mathematics / Cognitive Science Minor

(Expected) June 2027

- **Cumulative GPA: 4.0/4.0**, Provost Honors (Dean's List), Eta Kappa Nu – IEEE (Top 20%)

Coursework: Data Structures and Algorithms, Advanced Linear Algebra, Statistical Methods, Intro to Data Visualization

RESEARCH & EXPERIENCE

Halicioğlu Data Science Institute, UC San Diego

La Jolla, CA

Instructional Assistant

January 2025 - Present

- Held weekly office hours to assist a class of ~370 students with programming in Python and coursework, covering key computational topics such as recursion, object-oriented programming, and data structures.
- Provided individualized feedback during one-on-one reviews, aiding with conceptual understanding and debugging.

Data Science Student Society

La Jolla, CA

Sponsorship Chair

May 2024 – Present

- Assisted the VP External in maintaining strong relationships with industry partners such as Palantir and ASML, raising \$1700+ in sponsorships for the organization.
- Hosted ~12 networking/technical events with new industry partners, laying the groundwork for future partnerships.

Undergraduate-Graduate Research Lab, UC San Diego

La Jolla, CA

Research Assistant

April 2024 – November 2024

- Conducted a comprehensive literature review on multiproduction networks and trade credit, analyzing over 10 research papers to gain insight on firm-to-firm financial and production linkages
- Supported the development of a static model using Python with over 100,000 transaction records from Chilean firms.

PROJECTS

N-Gram Language Model Optimization

Machine Learning/Natural Language Processing (NLP)

- Built an N-Gram language model for text generation, implementing tokenization, n-gram probability estimation, and sampling, improving sentence coherence by 35%.
- Developed an efficient text generation pipeline by optimizing n-gram computation, reducing model perplexity by 25%, showcasing expertise in classical NLP techniques.

MTS Bus Network Analysis and Visualization

Data Processing/Statistical Modeling

- Analyzed and visualized San Diego Metropolitan Transit System (MTS) data from 100+ bus routes and 2500+ bus stops using GeoPandas and Plotly to map bus networks, optimize travel times, and visualize transit efficiency.
- Simulated and analyzed waiting times for 1000 passengers, revealing 5% difference between actual passenger wait times and the expected Poisson process, enhancing insights into optimizing public transit.

Credit Card Fraud Detection

Data Science/Machine Learning Project

- Developed a credit card fraud detection system using Logistic Regression and Neural Networks with Scikit-learn and TensorFlow, achieving 92% accuracy in classifying fraudulent transactions.
- Engineered and analyzed key transaction features (e.g., time, amount) to uncover patterns indicative of fraud, processing and evaluating over 280,000 transaction records to enhance detection capabilities.

SKILLS

Programming: Python (Pandas, Scikit Learn, TensorFlow), Java, SQL, MATLAB, Web Scraping, Object-Oriented Programming

Soft Skills: Communication, Leadership, Collaboration, Detail Oriented, Critical Thinking